

ITA0610 - Machine Learning

For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Experiment: Candidate Elimination Algorithm

Aim

To implement and demonstrate the **Candidate Elimination Algorithm** in Python using a **CSV file** and obtain the set of all hypotheses that are consistent with the given training examples.

Algorithm

1. Import the required Python libraries.
2. Read the training dataset from a CSV file.
3. Initialize the **Specific Hypothesis (S)** with the first positive training example.
4. Initialize the **General Hypothesis (G)** with the most general hypothesis.
5. For each training example:
 - If the example is **Positive (Yes)**:
 - Update the Specific Hypothesis by generalizing it where necessary.
 - Remove inconsistent hypotheses from the General Hypothesis.
 - If the example is **Negative (No)**:
 - Specialize the General Hypothesis to exclude the negative example.
6. Repeat until all training examples are processed.

7. Display the final Specific Hypothesis (S) and General Hypothesis (G)

PROGRAM:

```
import pandas as pd
```

```
import numpy as np
```

```
data = pd.read_excel("trainingdata.xlsx")
```

```
print("Training Data:\n")
```

```
print(data)
```

```
concepts = np.array(data.iloc[:, :-1])
```

```
target = np.array(data.iloc[:, -1])
```

```
def learn(concepts, target):
```

```
    specific_h = concepts[0].copy()
```

```
    general_h = [["?" for i in range(len(specific_h))]
                  for j in range(len(specific_h))]
```

```
    for i, h in enumerate(concepts):
```

```
        if target[i] == "Yes":
```

```
for x in range(len(specific_h)):
```

```
    if h[x] != specific_h[x]:
```

```
        specific_h[x] = "?"
```

```
        general_h[x][x] = "?"
```

```
elif target[i] == "No":
```

```
for x in range(len(specific_h)):
```

```
    if h[x] != specific_h[x]:
```

```
        general_h[x][x] = specific_h[x]
```

```
    else:
```

```
        general_h[x][x] = "?"
```

```
general_h = [g for g in general_h if g != ["?"] * len(specific_h)]
```

```
return specific_h, general_h
```

```
specific, general = learn(concepts, target)
```

```
print("\nFinal Specific Hypothesis:")
```

```
print(specific)
```

```
print("\nFinal General Hypothesis:")
```

```
for g in general:
```

```
    print(g)
```

OUTPUT:

```
print(specific)

print("\nFinal General Hypothesis:")
for g in general:
    print(g)
```

Training Data:

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

Final Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']

Result

The **Candidate Elimination Algorithm** was successfully implemented and executed using the given training dataset. The algorithm generated the **Specific Hypothesis (S)** and **General Hypothesis (G)**, representing the set of all hypotheses consistent with the training examples. The final specific hypothesis obtained was:

['Sunny' 'Warm' '?' 'Strong' '?' '?']

The final general hypotheses obtained were:

['Sunny', '?', '?', '?', '?', '?']

['?', 'Warm', '?', '?', '?', '?']

Hence, the Candidate Elimination algorithm successfully identified the version space consistent with the given training data.